

Piyush Bindua

AI ENGINEER · APPLIED AI | GENAI · RAG · LLM APPS

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Skills

GenAI & LLMs: RAG (hybrid, GraphRAG, reranking), LangChain / LangGraph, MCP, prompt engineering, embeddings, fine-tuning (LoRA / QLoRA)

LLMOps & Eval: RAGAS, DeepEval, LangSmith / Langfuse

Data Stores: Qdrant, pgvector, Neo4j, PostgreSQL, Redis, FAISS

Languages: Python, SQL

Backend: FastAPI, REST, OAuth 2.1

Models: Llama, Gemini, Claude, GPT ·

Experience

AI Engineer — Quasi Intelligence

July 2026 – Sep 2026

Indore, India

Quasi RAG — on-premise GraphRAG assistant for technical documentation

LangGraph · Qdrant · BM25 · Kuzu / Neo4j · weighted RRF · FlashRank · Ollama / Gemma 3 · FastAPI

- Built an air-gapped GraphRAG system to allow engineers to query datasheets and schematics where documents can't leave the network - Ollama/Gemma 3, FastEmbed, and ONNX rerankers running fully on-premise with zero external API calls
- Fused dense (Qdrant), lexical (BM25), and graph (Kuzu/Neo4j) retrievers in parallel via weighted Reciprocal Rank Fusion with a cross-encoder reranker, adding Louvain community summaries for whole-document questions and a query router that reweights toward lexical retrieval for component references, where dense embeddings smear short distinctive tokens.
- Implemented a three-layer hallucination defence — a rerank-score floor that refuses without invoking the LLM, a strict prompt mandating inline citations, and a post-hoc verifier auditing each claim against its cited passage — orchestrated as a LangGraph state machine with per-thread memory, served via FastAPI with click-through PDF.js citations.

Deloitte, Data Analytics Virtual Experience Program (Forage)

Sep 2025 – Dec 2025

- Completed a **data analytics and forensic technology** job simulation based on real-world business scenarios.
- Conducted **EDA, data cleaning, and transformation** using **Python and SQL** to identify patterns and insights.
- Built **Tableau and Power BI** dashboards to analyze and present KPIs for data-driven decision making.

PROJECTS

Fire & Smoke Detection System

Python · YOLO11n · OpenCV · Google Colab

- Developed a **Fire & Smoke Detection** system using **YOLO11n**, training on **4,304 images** and achieving **89.4% mAP@50** and **62.8% mAP@50-95**.
- Prepared and annotated image datasets in **YOLO format**, including data organization and preprocessing for object detection model training.
- Trained and fine-tuned **YOLO models** using **Python, OpenCV, and Google Colab**, and evaluated performance using **Precision, Recall, IoU, and mAP**.
- Processed images and video for computer vision applications, working on **object detection, inference, and model evaluation**.
- Used **Git and GitHub** for version control, project development, and documentation.

Multi-Tenant RAG System (AI-Powered Knowledge Retrieval Platform)

Technologies: Node.js, TypeScript, Express.js, PostgreSQL, pgvector, OpenAI

- Built a **multi-tenant RAG platform** that allows organizations to upload, manage, and query private documents using AI-powered semantic search and document retrieval.
- Designed **tenant-isolated data access** with Node.js, TypeScript, PostgreSQL, and pgvector to keep each organization's knowledge base securely separated.
- Developed an end-to-end **RAG pipeline** covering document ingestion, text processing, vector embeddings, similarity search, and context-aware response generation.
- Built **REST APIs** with Express.js and added AI guardrails to handle prompt injection, low-confidence retrieval, and unauthorized information exposure.

Education

B.Tech, Computer Science — Acropolis Institute of Technology and Research

Sep 2021 – Jun 2025

Indore, Madhya Pradesh

Senior Secondary (12th Grade)

Year 2021

School for Excellence, Chhatarpur, Madhya Pradesh — Percentage: 82.6%

Secondary (10th Grade)

Year 2019

School for Excellence, Chhatarpur, Madhya Pradesh — Percentage: 82%

Achievements

- Deloitte Data Analytics (Forage, 2025)
- Cisco Cybersecurity (2024)
- Salesforce AI Associate (2024)